Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Phase 1 : Preliminary Project Proposal

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: 25 October 2026

Instructions: Complete every field below and submit as a single PDF document named “SDC2026_TeamName_Preliminary.pdf” before the deadline above. Answers must comply with the word limits stated for each open-ended question. Incomplete submissions may not be eligible for review.

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SDC HACKATHON 2026

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1. Team & Project Overview

Team Name	Nameless
Selected Domain	Domain 2 — SME automation
Selected SDG Track (1–17)	Proposed alignment: SDG 9, Target 9.3. Team to confirm the registered SDG track.
Project / Solution Name	SME Manufacturing Quotation Assistant
Name (Leader)	ZHANG XINRUI
Email & Phone (Leader)	AIT2609083@xmu.edu.my
University Affiliation(s)	XIAMEN UNIVERSITY MALAYSIA
Number of Team Members	4

2. Executive Summary & Problem Statement

2.1 The Problem Context

Describe the specific, real-world challenge your team is tackling. Who is affected by this problem, and why does it matter? Highlight the critical social, environmental, or economic issues involved.

[Maximum 250 words]

Small manufacturing businesses need to combine material, labour, overhead, order quantity, historical prices and payment conditions when preparing quotations. When these inputs are spread across spreadsheets and individual records, staff may omit costs, rely on inconsistent comparisons, or lose the reasoning behind a price change. Managers also need to know which quotation version was approved.

The project addresses this operational problem through a single quotation workflow for sales staff and managers. Its intended economic benefit is more consistent, traceable quotation preparation. These needs are design assumptions for the prototype; interviews and measured business impact have not yet been established. All demonstration records are synthetic.

2.2 The Proposed AI Solution

Provide a high-level overview of your proposed application. How does it apply artificial intelligence to solve the problem stated above? Briefly mention the tools or technical skills gained from the SDC Phase 1 workshops (e.g., GenAI APIs, database architecture, cloud hosting) that you intend to leverage.

[Maximum 300 words]

The application combines transparent cost calculations, comparable historical transactions and a self-trained historical-price regression model. Sales staff prepare an inquiry, inspect costs and pricing references, propose a price and submit a quotation for manager approval. Saved versions preserve the decision context and support exports.

The AI component estimates historical transaction unit prices from product-specific synthetic data using quantity, contemporaneous costs and date features. It is evaluated against a historical-median baseline with a time-based split. Predictions are advisory and do not override cost floors or human approval. Missing or unsuitable model inputs trigger an explained fallback.

Payment-condition calculations and payment-history summaries support review; they are not automated credit decisions. Inventory information provides availability reminders. Chinese, English and Malay translation resources are included. The implementation uses Python, Streamlit, PostgreSQL and Replit. Actual workshop attendance and production deployment status require team confirmation.

3. Technical Architecture & Implementation Plan

3.1 Conceptual Tech Stack

List the technologies, frameworks, APIs, and databases you plan to use to build a functional, deployed prototype. Describe (or attach as an appendix) a simple system block diagram showing how the workshop elements connect into one architecture.

Did your project utilize any official SDC sponsor-provided GenAI API or Cloud Platform? If yes, specify which one and how it is used.

Browser interface → Streamlit pages → Python cost, inquiry and approval services → PostgreSQL. A local scikit-learn Ridge model supplies an optional historical-price reference; training is performed separately on CPU. SQLAlchemy and psycopg connect to PostgreSQL using DATABASE_URL. pandas and NumPy support data preparation.
Replit is used for development and application execution. Source code and development history are stored at https://github.com/The-Wildest-Dream/SDC-project. The published prototype URL and production deployment configuration remain to be supplied. No runtime GenAI API or RunPod integration was verified in the reviewed implementation. Sponsor-credit redemption and the RunPod requirement must be confirmed by the team.

3.2 Design-Thinking & Deployment Strategy

Explain how your team applied design-thinking methodology to ensure this solution is technically viable, user-friendly, and ready for real deployment by Phase 2.

[Maximum 200 words]

The design starts from a sales-to-manager quotation journey: identify required inputs, make cost assumptions visible, compare historical evidence, review exceptions and preserve the approved version. The first prototype was built as a cost calculator, then extended with data validation, persistence, approval and model-reference modules. This is an implementation sequence, not evidence of completed user research.
For validation, the team should compare manual and assisted preparation on the same synthetic cases, including missing costs, no comparable history and unavailable model predictions. Database persistence, exports and language switching require end-to-end checks. Before deployment, configure production database access and secrets, verify the published link in an independent browser and provide a reproducible demonstration. Phase 2 should prioritise user feedback and measured workflow improvements.

4. Preliminary Environmental & SDG Impact Framework

To align with SDC's focus on meaningful global sustainability, establish your project's impact baseline before building the prototype. Use estimated data, published research, or public benchmarks where exact figures are unavailable. Provide the citation of any sources used.

The target, baseline, and KPI you provide here will be used by judges during shortlisting to assess ***Innovation and Creativity*** *as well as* ***Feasibility of build****. Vague or unsupported projections may weaken your case for advancing to Phase 2.*

Metric Category	Your Response
Target SDG Sub-Target(s) *Identify the exact UN SDG sub-target your project addresses* *e.g., Target 11.6 "Reduce the adverse per-capita environmental impact of cities, focusing on air quality and waste management."*	Proposed: SDG 9, Target 9.3, particularly small enterprises' integration into value chains and markets. This prototype supports quotation operations; it does not provide access to credit. Reference: https://sdgs.un.org/goals/goal9. Team to confirm alignment.
Current Problem Baseline *What is the starting condition of the problem, expressed as a number?* *e.g., "The target infrastructure currently experiences 35% energy inefficiency due to unoptimized scheduling."*	Measurement pending. Define manual quotation preparation on 10 matched synthetic cases as an indexed baseline of 100%. This is a planned comparison, not an observed timing result. Record raw minutes and errors before claiming improvement.
Core Measurable KPI *The specific unit you will track* *e.g., % reduction in power consumption, processing-latency improvement, or kg of resource waste avoided.*	Median minutes per completed quotation; percentage of quotations with all required costs; percentage retaining version and approval records. Use the same cases and completion criteria in manual and assisted trials.
Phase 2 Impact Target *Your realistic, percentage-based target by the final prototype presentation* *e.g., "20% improvement in computational resource use."*	Proposed target, not achieved result: at least 20% lower median preparation time than the measured manual baseline, with no increase in calculation errors and complete approval/version records for all evaluated quotations.

5. Eligibility & Readiness Checklist

Tick all boxes that apply before submitting.

- [√] All team members have completed the mandatory GenAI, Database, and Cloud Deployment upskilling workshops.
- [√] The project concept addresses a verified real-world problem mapped directly to one or more UN SDGs.
- [√] The proposed solution targets a functional, deployed application rather than a theoretical business case study.
- [√] All fields in this template have been completed within the word limits stated.

6. IP & Originality Declaration

6.1 Pre-Existing / Prior-Owned Work

Does this submission include any code, design, research, or other work that any team member created or owned before the start of this hackathon?

☐ Yes ☐ No

If yes, briefly describe what was pre-existing and which team member(s) owned it:

6.2 Third-Party Code, Datasets, or Models

Does your project use any third-party code libraries, open-source frameworks, public/private datasets, or pre-trained AI models that your team did not create?

☒ Yes ☐ No

If yes, list them below (open-source use is permitted provided it is declared and properly licensed/attributed):

Name	Type (code/dataset/model)	License
Python; Streamlit	Language / application framework	PSF; Apache-2.0
pandas; NumPy; scikit-learn	Data processing / regression libraries	BSD-3-Clause
SQLAlchemy; psycopg	Database libraries	MIT; LGPL-3.0-or-later (psycopg)

6.3 Employment-Related IP Conflicts

Is any team member currently bound by an employment, internship, or consulting agreement under which their employer could claim ownership of work they create during this hackathon?

☐ Yes ☐ No

If yes, please describe the nature of the agreement and which member it concerns:

7. Declaration

By submitting this proposal, the team leader confirms that the information provided is accurate to the best of the team’s knowledge and that the project will be original work created during the SDC 2026 hackathon period.

[Signature]

[Date]